

SORT Function in Excel

1. What is SORT?

The SORT function returns a sorted copy of a list or table, in ascending or descending order. It solves the problem of putting data in a useful order, such as names from A to Z or sales from highest to lowest. A Data Analyst uses it to build ranked lists and ordered reports that update automatically when the source data changes, without changing the original table.

2. Syntax

`=SORT(array, [sort_index], [sort_order], [by_col])`

Argument	Required?	What it accepts
array	Required	The range or table to sort (for example A2:C7)
sort_index	Optional	The number of the column to sort by, counted within the array. The default is 1 (the first column).
sort_order	Optional	1 for ascending, smallest to largest or A to Z (default). -1 for descending.
by_col	Optional	FALSE (default) sorts rows. TRUE sorts columns from left to right. Beginners rarely need TRUE.

Text sorts alphabetically and numbers sort from smallest to largest. In A2:C7, a `sort_index` of 3 means the third column of the array, which is column C.

3. Simple Example

Sort a list of products from A to Z:

	A	B	C
1	Product		Sorted Product
2	Orange		Apple
3	Apple		Banana
4	Mango		Grape
5	Banana		Kiwi
6	Grape		Mango
7	Kiwi		Orange

Formula in C2 only: `=SORT(A2:A7, 1, 1)` (the heading in C1 is typed)

The list has one column, so the sort index is 1. The sort order 1 means ascending. Excel sorts the six products from A to Z and spills them into C2 to C7. **Result: Apple, Banana, Grape, Kiwi, Mango, Orange**

4. Another Example

Rank employees by sales, highest first:

	A	B	C	D	E	F	G
1	Name	Dept	Sales		Name	Dept	Sales
2	Ayesha	IT	850		Nabila	IT	910
3	Rahim	HR	620		Ayesha	IT	850
4	Nabila	IT	910		Fahim	Finance	780
5	Fahim	Finance	780		Sadia	IT	660
6	Sadia	IT	660		Rahim	HR	620
7	Rafi	Sales	540		Rafi	Sales	540

Formula in E2 only: `=SORT(A2:C7, 3, -1)` (the headings in E1:G1 are typed)

Excel sorts the whole table by its 3rd column (Sales). The sort order -1 means descending, so the highest sales come first. Each name stays with its own department and sales. **Result: Nabila at the top, Rafi at the bottom**

5. When to Use It

- Rank top performers, products, or regions by sales or score
- Create alphabetical lists of names, products, or customers
- Keep a report in order automatically as the data changes
- Sort the results of other functions, such as FILTER or UNIQUE
- Show a sorted view of the data while keeping the original table untouched

6. Important Things to Remember

- SORT is a dynamic array function. One formula fills many cells, so the cells for the result must be empty, or Excel shows `#SPILL!`.
- The original data is not changed. SORT returns a sorted copy and updates when the source changes.
- `sort_index` is the position of the column inside the array, not the column letter on the sheet.
- `sort_order` must be 1 (ascending) or -1 (descending). Any other number gives `#VALUE!`.
- Select the whole table in `array` so that all columns move together with the sorted column.
- SORT is available in Excel 2021, Excel 2024, and Microsoft 365, but not in Excel 2019 or earlier.

7. Common Mistakes

Mistake: Selecting only the column to be sorted, so the other columns do not follow it. **Fix:** Select the whole table in `array` and use `sort_index` to choose the column.

Mistake: Using the sheet column number for `sort_index`. For B2:D7, a `sort_index` of 3 sorts by column D, not column C. **Fix:** Count from the first column of the array.

Mistake: Including the header row in the array, so the heading is sorted into the data. **Fix:** Start the range at the first data row.

Mistake: Using 0 or 2 for `sort_order`, which gives `#VALUE!`. **Fix:** Use 1 for ascending or -1 for descending.

8. Data Analyst Tip

To sort by two columns, give array constants for the second and third arguments. For example, `=SORT(A2:C7, {2,3}, {1,-1})` sorts by Department A to Z, then by Sales high to low within each department. You can also wrap SORT around other functions, such as `=SORT(UNIQUE(B2:B7))`, for a sorted list of unique values.